

Institutional NAC Pre-Vetting & Risk Mitigation Framework

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Project: PoC 21 - Portfolio Optimization & Risk Mitigation

Tech Stack: Python | Streamlit | LangChain (RAG) | ChromaDB | Groq (Llama 3.3-70B)

Project Vision

In institutional banking, the New Activity (NAC) process is often a strategic bottleneck. This Proof of Concept (PoC) demonstrates how Retrieval-Augmented Generation (RAG) can shift governance from a reactive "No" to a proactive "Pre-Audit."

The tool acts as a Digital First-Responder, allowing Managing Directors to vet partnership proposals against complex internal policies in seconds, identifying regulatory "red flags" and financial misalignments before they reach the committee.

Executive Summary

Strategic Intent: To transform the 'New Activity' (NAC) process from a reactive bottleneck into a proactive strategic filter.

The Problem

Institutional deal teams often spend weeks preparing partnership proposals, only to face "late-stage rejection" due to governance or risk misalignments. This lag increases operational friction and slows the institutional deal pipeline.

The Solution

To eliminate the bottle neck, this PoC proposes the use of a Retrieval-Augmented Generation (RAG) framework that acts as a "Digital First-Responder." By cross-referencing multi-million-dollar deal proposals against internal bank policies in real-time, the tool identifies regulatory 'red flags' and provides an instant go/no-go risk assessment of a partnership proposal.

Key Value Drivers

- **Instant Visibility:** A dynamic Risk Heatmap provides immediate clarity on Financial, Regulatory, and Operational exposure.
- **Explainable AI (XAI):** Each finding is backed by an indisputable audit trail, quoting exact policy snippets from the internal knowledge base.
- **Operational Efficiency:** Drastically reduces committee review cycles by ensuring only "governance-ready" proposals move forward.

Key Features

- **Dual-Engine RAG:** Combines deep narrative audit analysis with a dedicated quantitative scoring engine.
- **Risk Exposure Heatmap:** A color-coded visual dashboard (Yellow-Orange-Red) for instant risk quantification.
- **Indisputable Audit Trail:** Every finding includes Source Snippets—direct quotes from internal governance PDFs to ensure 100% transparency.
- **Executive Export:** Generates professional, board-ready PDF audit reports for formal submission.
- **Scalable Knowledge Base:** Modular design allows for instant policy updates via simple PDF ingestion without model retraining.

Technical Architecture

- **Orchestration:** LangChain (RAG Architecture)
- **LLM:** Llama 3.3-70B (via Groq) for high-speed institutional reasoning.
- **Vector Database:** ChromaDB (Local persistent storage).
- **Embeddings:** HuggingFace all-MiniLM-L6-v2 (100% local/private processing).
- **Frontend:** Streamlit (Optimized for Managing Director-level UX).

Scalability Roadmap: From PoC to Production

- **Multi-Agent Review:** Deploy specialized "AI Agents" for specific risk domains (e.g., a "Legal Agent" and a "Financial Audit Agent") to provide deeper cross-functional analysis.
- **Enterprise Auth:** Integration with Okta/Azure AD for secure, role-based access to sensitive governance documents.
- **Audit Trail Logging:** Connecting the tool to an immutable database (e.g., SQL or Blockchain) to log every pre-audit for future regulatory reviews.
- **Feedback Loop:** Allowing Risk Officers to "thumbs up/down" AI findings to fine-tune the RAG retrieval precision over time.

Installation & Usage

1. Clone & Setup:

```
pip install -r requirements.txt
pip install fpdf2 seaborn matplotlib pandas
```



2. Launch Portal:

```
streamlit run app.py
```



3. **Analyze:** Enter your Groq API key, ingest your policy library (/data/governance_docs), and upload a proposal to generate your first audit.

Dashboard Strategy

The UI is optimized for a Managing Director's review, and uses a 1:2 layout ratio that prioritizes a narrow, high-impact Risk Dashboard on the left and a detailed, high-readability Policy Narrative on the right. The policy narrative can be downloaded into a .pdf file format.

User Guide: NAC Pre-Vetting Portal

Role: Self-Service Strategic Alignment & Risk Identification

1. Getting Started

- **Authentication:** Enter your Groq API Key in the sidebar. This connects the secure "reasoning engine" to the portal. (A Groq API Key may be obtained by accessing the GroqCloud Console (<https://console.groq.com/home>) From this page a user can create a free account using their e-mail address, existing Google e-mail, or GitHub credentials.
- **Knowledge Sync:** If new governance policies have been released, click "Update/Ingest Governance Docs." The system will scan the internal library and refresh its "memory" (this takes about 30–60 seconds).

2. Running an Assessment

- **Upload:** Drag and drop your Partnership Proposal (PDF or TXT) into the upload zone.
- **Analyze:** Click " Run Risk Assessment."
- **What happens next:** The AI performs two tasks. First, it searches the policy library for relevant rules. Second, it generates the detailed audit and calculates the risk scores.

****Analysis of Proposal against Governance Context****

The proposal, PROJECT NEPTUNE, submitted by FintechStream LLC, aims to establish a strategic partnership with National Bank Group to offer a co-branded "Instant Credit" product for small business owners. The proposal is analyzed against the National Bank Group New Activity (NAC) policy, which outlines specific rules and guidelines for new partnership activities.

****Non-Compliance with NAC Policy:****

1. ****Financial Viability (Rule 1.1):**** The proposal lacks a complete 3-year P&L forecast, which is a mandatory requirement. Although a draft is mentioned, it is not sufficient to meet the policy's requirements. Furthermore, the estimated 10% IRR in Year 1 is below the minimum threshold of 12% required by the policy, and no written waiver from the Divisional CFO is mentioned.

2. ****Data Sovereignty (Rule 3.2):**** The proposal plans to store client credit scores on FintechStream's proprietary servers in a Caribbean-based data center, which may not be a Tier-1 jurisdiction. This raises concerns about data sovereignty and security.

3. ****Risk Mitigation and Crisis Exit Strategy:**** The proposal lacks a formal "Crisis Exit Strategy," which is a common reason for proposal rejection, as noted in the Historical Note. Although FintechStream uses a proprietary AI model to assess creditworthiness, the absence of a clear risk mitigation plan for interest rate volatility is a significant concern.

****Reputational Risk (Rule 2.4):**** The proposal does not explicitly mention partnership with entities involved in